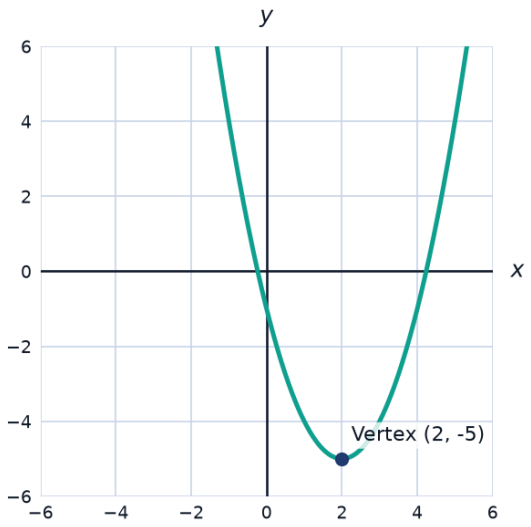
## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the quadratic function $f(x) = x^2 + bx + c$. The graph of $f(x)$ is shown below. The vertex of the parabola is at $(2, -5)$ and it passes through the point $(0, -1)$. Determine the values of $b$ and $c$.

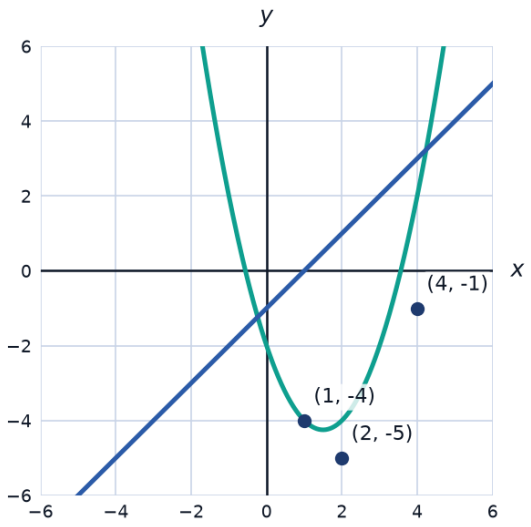



**2** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The graph of the quadratic function $f(x) = 2x^2 - kx + 5$ passes through the point (3, 14). Find the value of $k$.

**3** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

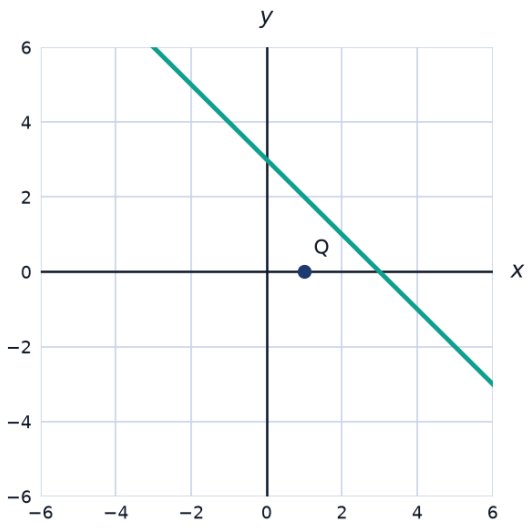
The diagram shows the graph of the function $f(x) = ax^2 + bx + c$. The graph passes through the points $(1, -4)$, $(2, -5)$, and $(4, -1)$. Determine the value of $a$.




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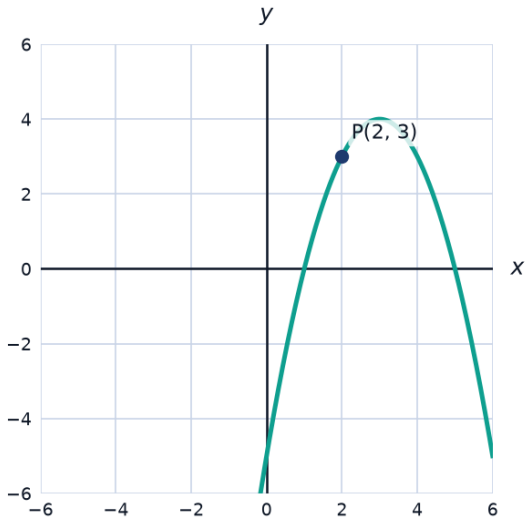
**4** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the two functions $f(x) = \log_2(x)$ and $g(x) = 3 - x$. The graphs of $f(x)$ and $g(x)$ intersect at a point P, and $f(x)$ intersects the x-axis at point Q. A line segment is drawn from Q to a point R on the graph of $g(x)$ such that QR is perpendicular to the x-axis. Find the coordinates of the point R.




**5** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

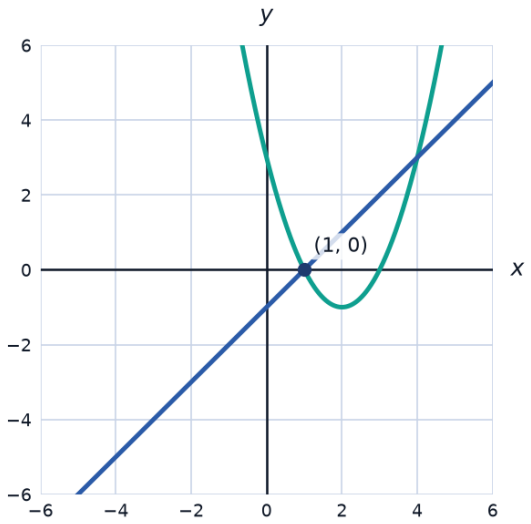
Consider the function $f(x) = -x^2 + 6x - 5$. The graph of this function is shown below. A tangent line to the graph at point $P(2, 3)$ intersects the x-axis at point $Q$. Determine the coordinates of point $Q$.




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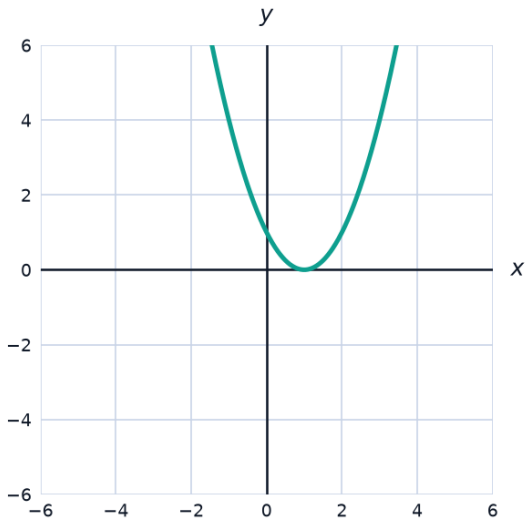
**6** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the function $f(x) = x^2 - 4x + 3$ and the line $g(x) = x - 1$. The graph of these functions is shown below. Find the coordinates of the point where the line intersects the parabola.




**7** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the function $f(x) = x^2 - 2x + 1$. Determine the y-coordinate of the vertex of the graph of $y = f(x)$.

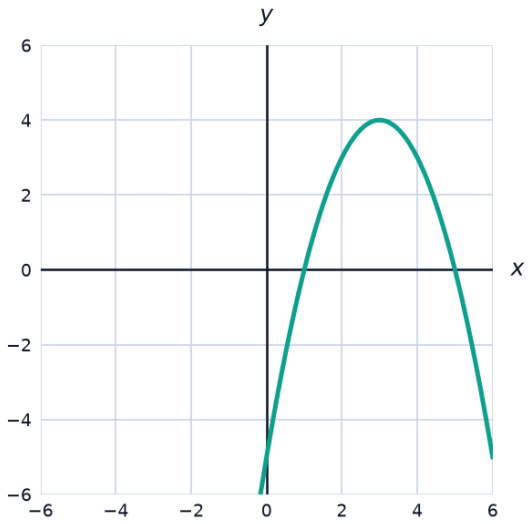



## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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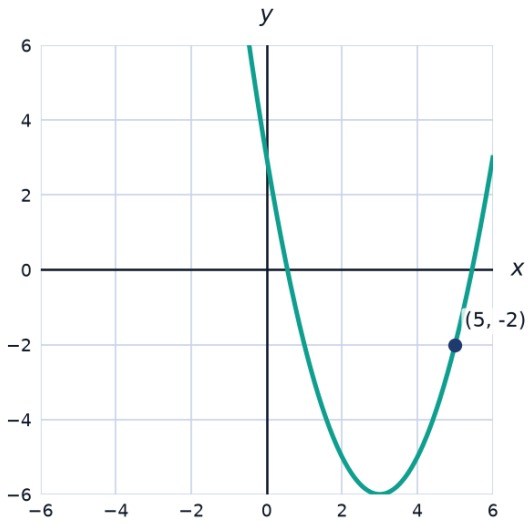
**8** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the function $f(x) = -x^2 + 6x - 5$. Determine the coordinates of the vertex of the graph of $y = f(x)$.




**9** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The graph of the function $f(x) = x^2 - 6x + k$ is shown below. The graph passes through the point $(5, -2)$. Find the value of $k$.




**10** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the function $f(x) = e^{-x^2}$ and the line $y = k$. The graph of $y = f(x)$ is concave up for $x \in (-\infty, -a) \cup (a, \infty)$ and concave down for $x \in (-a, a)$. Find the exact value of $a^2$.

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**11** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The depth of water $h(t)$ in meters at a harbor entrance $t$ hours after midnight is modelled by $h(t) = 3.5\cos\left(\frac{\pi t}{6}\right) + 6.5$ for $0 \le t \le 24$. (a) State the maximum and minimum depths of water and their times of occurrence. (b) Find the period of the tidal oscillation. (c) A cargo ship requires a minimum water depth of 8.25 meters to safely enter the harbor. Find the total number of hours during the first 12 hours of the day when the ship can safely enter.

**12** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the function $f(x) = x^3 - 6x^2 + 5$. Find the exact value of the sum of the x-coordinates of the local maximum and local minimum points of the graph of $y = f(x)$.

**13** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The function $f(x) = \frac{2x-5}{x+3}$ is defined for $x \neq -3$. (a) Find the coordinates of the axes intercepts of the graph of $y = f(x)$. (b) Write down the equations of the vertical and horizontal asymptotes. (c) Find an expression for $f^{-1}(x)$ and state its domain and range.

**14** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In the triangle ABC shown, point D lies on AB and point E lies on AC such that DE is parallel to BC. Given that AD = 6, DB = 4, and the area of triangle ADE is 27 square units, find the area of triangle ABC.

## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

**15** [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 AI SL — Specific Topics

The box plot below shows the distribution of scores for a class on a recent statistics quiz. The five-number summary is given by minimum, first quartile (Q1), median, third quartile (Q3), and maximum. If the interquartile range (IQR) is 25, and the median score is 70, find the value of Q1.

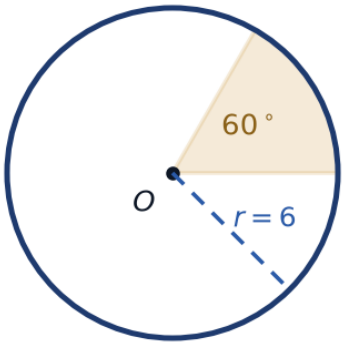
**16** [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Functions

Evaluate the definite integral $\int_0^{\sqrt{7}} \frac{4x}{\sqrt{x^2+9}}\,dx$ using the substitution $u = x^2 + 9$.

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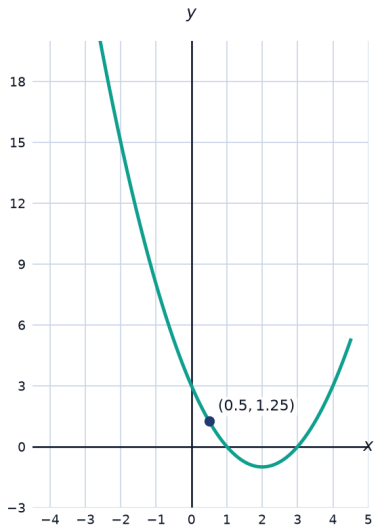
**17** [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Geometry & Trigonometry

Solve the trigonometric equation $2\sin^2\theta + 3\cos\theta - 3 = 0$ for $0 \le \theta \le 2\pi$, giving exact radian answers.




**18** [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Statistics & Probability

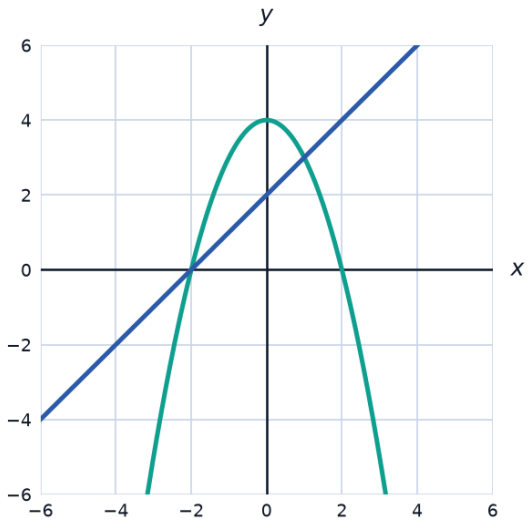
Consider the function $f(x) = x^2 - 4x + 3$ and the function $g(x) = x - 1$. The graphs of $y = f(x)$ and $y = g(x)$ intersect at points $A$ and $B$. One of these intersection points is $(1, 0)$. Find the coordinates of the other intersection point.




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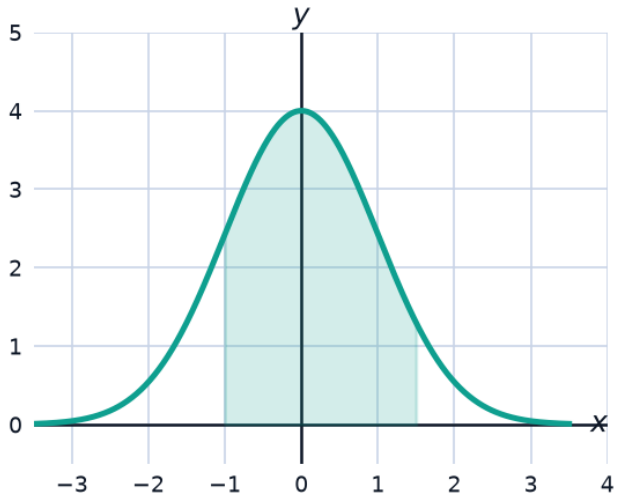
**19** [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.01 Calculus

Solve the logarithmic equation $\log_3(x+5) + \log_3(x-1) = 2 + \log_3(2)$ for all real values of $x$.




**20** [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 AI SL — Specific Topics

A discrete random variable $X$ follows a binomial distribution $X \sim B(15, 0.35)$. (a) Find $P(X = 5)$. (b) Find $P(X \geq 4)$. (c) Calculate the mean $E(X)$ and standard deviation $\sigma$ of $X$.

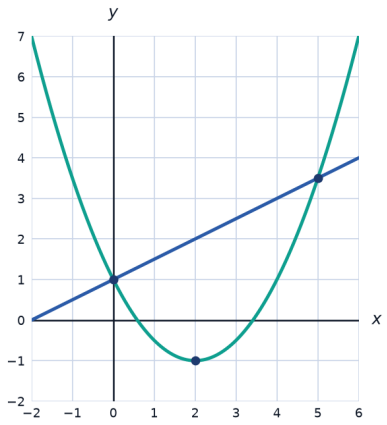



## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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**21** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 2x + 3$ in solve by factorisation, evaluate $f(2)$.




**a.** 7

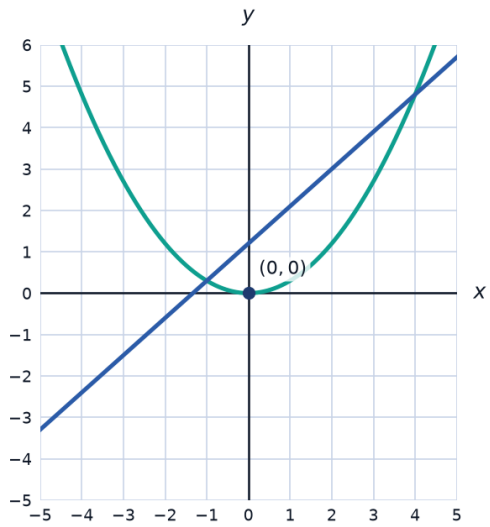
**b.** 9

**c.** 4

**d.** 11

**22** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.




**a.** $x = 2$ and $x = 5$

**b.** $x = 2$ and $x = 4$

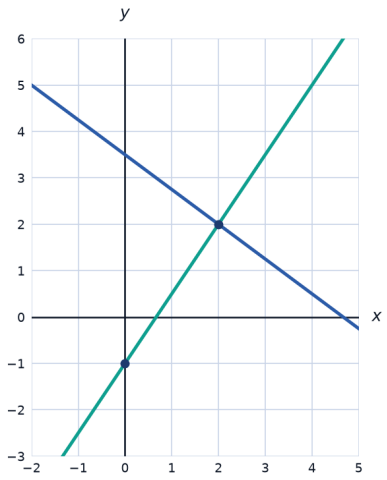
**c.** $x = -2$ and $x = -4$

**d.** $x = 6$ and $x = 8$

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**23** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 3x + 4$ in solve quadratics by completing the square, evaluate $f(2)$.




**a.** 10

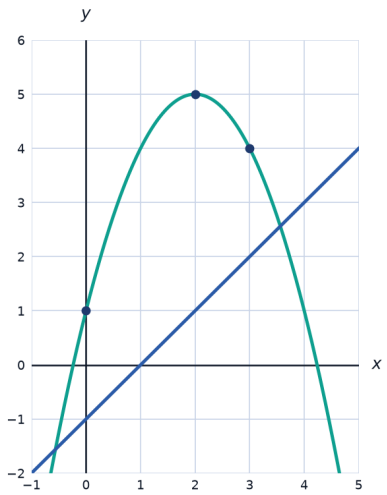
**b.** 13

**c.** 6

**d.** 14

**24** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 4x + 5$ in solve using the quadratic formula, evaluate $f(2)$.




**a.** 13

**b.** 17

**c.** 8

**d.** 17

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**25** [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for solve by factorisation consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

**a.** 14.5

**b.** 15.5

**c.** 16.5

**d.** 18.5

**26** [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for solve quadratics by completing the square consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

**a.** 15.5

**b.** 16.5

**c.** 17.5

**d.** 19.5

**27** [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for discriminant of a quadratic consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

**a.** 17.5

**b.** 18.5

**c.** 19.5

**d.** 21.5

**28** [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 5x + 6$ in discriminant of a quadratic, evaluate $f(4)$.

**a.** 26

**b.** 31

**c.** 20

**d.** 30

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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**29** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for solve quadratics using various methods consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

**a.** 18.5

**b.** 19.5

**c.** 20.5

**d.** 22.5

**30** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 6x + 7$ in solve quadratics using various methods, evaluate $f(4)$.

**a.** 31

**b.** 37

**c.** 24

**d.** 35

**31** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 2x + 8$ in graph basic quadratics, evaluate $f(3)$.

**a.** 14

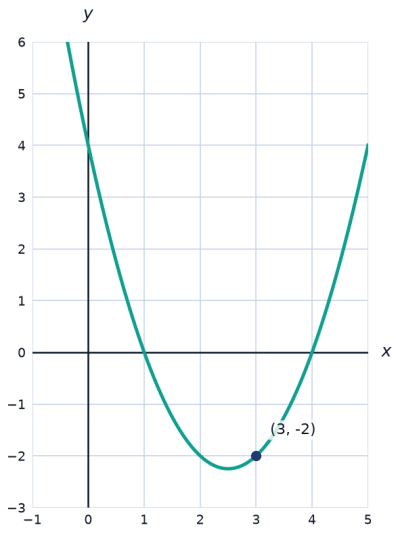
**b.** 16

**c.** 6

**d.** 18

**32** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

The graph of the quadratic function $f(x) = x^2 - 5x + k$ is shown below. The graph passes through the point $(3, -2)$. Find the value of $k$.




**a.** 2

**b.** 5

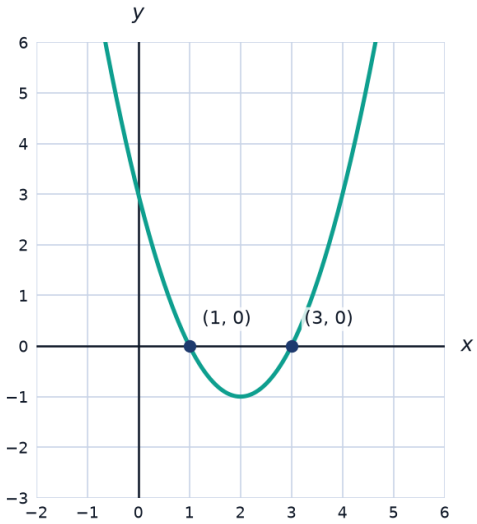
**c.** 7

**d.** 9

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**33** **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

The graph of the quadratic function $f(x) = x^2 - 4x + 3$ is shown below. What are the coordinates of the x-intercepts of the function?




**a.** (1, 0) and (3, 0)

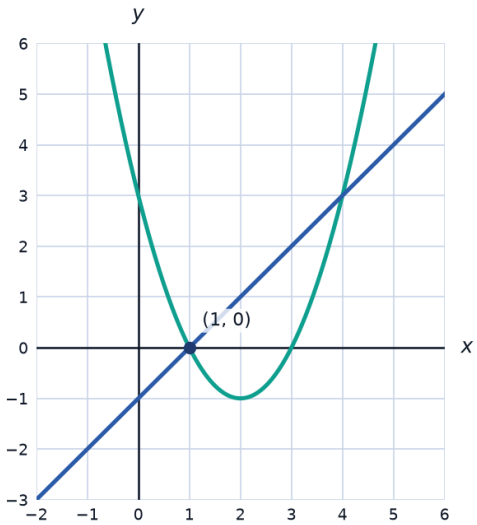
**b.** (0, 1) and (0, 3)

**c.** (-1, 0) and (-3, 0)

**d.** (0, -1) and (0, -3)

**34** **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

The graph of the quadratic function $f(x) = ax^2 + bx + c$ is shown below. It passes through the points $(0, -3)$, $(1, 0)$, and $(-1, -4)$. Find the value of $c$.




**a.** -3

**b.** 3

**c.** -1

**d.** 1

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**35** [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

The graph of the quadratic function $f(x) = x^2 - 6x + 5$ is shown below. Find the value of the y-intercept.




**a.** 5

**b.** -5

**c.** 0

**d.** 1

**36** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A particle's position on the x-axis at time $t$ (in seconds) is given by the function $x(t) = t^3 - 6t^2 + 9t + 1$. The particle starts at $t = 0$. Determine the time intervals during which the particle is moving in the negative direction and calculate the total distance traveled by the particle during the first 4 seconds.

**a.** Total distance = 10 units

**b.** Total distance = 12 units

**c.** Total distance = 14 units

**d.** Total distance = 16 units

**37** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Consider a cubic polynomial $P(x) = ax^3 + bx^2 + cx + d$. Given that $P(1) = 10$, $P(2) = 27$, $P(3) = 58$, and $P(4) = 115$, determine the value of $a$.

**a.** 1

**b.** 2

**c.** 3

**d.** 4

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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**38** [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Statistics & Probability

Consider a geometric progression where the sum of the first $n$ terms is given by $S_n = \frac{3(2^n - 1)}{2}$. A second sequence, denoted by $a_n$, is defined such that $a_n = S_n - S_{n-1}$ for $n \geq 2$, and $a_1 = S_1$. Find the value of $a_5 - a_3$.

**a.** 24

**b.** 18

**c.** 48

**d.** 36

**39** [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Calculus

The histogram below shows the number of hours 100 students spent studying for an exam. The number of students who studied for between 0 and 5 hours is represented by a bar with a height of 30. Determine the number of students who studied for between 5 and 10 hours.

**a.** 20

**b.** 25

**c.** 30

**d.** 35

**40** [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 6.01 AI SL — Specific Topics

Consider the equation $\log_x(81) = 4$. If $x$ is a positive real number, find the value of $x$.

**a.** 3

**b.** $\sqrt{3}$

**c.** 9

**d.** $2\sqrt{3}$

**41** [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Functions

The graph of the function $f(x) = \sqrt{x - 1} + 2$ is shown below. What is the value of $f(5)$?

**a.** 2

**b.** 3

**c.** 4

**d.** 5

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**42** [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 3.01 Geometry & Trigonometry

How many different 4-digit codes can be formed using the digits 1-9 without repetition, if the code must be even?

**a.** 1344

**b.** 3024

**c.** 504

**d.** 672

**43** [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 4.01 Statistics & Probability

Consider the function $f(x) = 2x - 3$. Determine the value of $f(3)$.

**a.** 3

**b.** 6

**c.** 9

**d.** 0

**44** [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Calculus

How many ways can 4 different books be arranged on a shelf?

**a.** 24

**b.** 12

**c.** 16

**d.** 8

**45** [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.01 AI SL — Specific Topics

Consider the function $f(x) = 2x + 5$. What is the value of $f(3)$?

**a.** 11

**b.** 8

**c.** 10

**d.** 16

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